MATH 8090:
Spectral Analysis of Time Series I

Readings: CC08 Chapters 13–14; BD16 Chapter 4; SS17 Chapters 4.1–4.4

Whitney Huang



School of
MATHEMATICAL AND
STATISTICAL SCIENCES
Clemson University

Week 11

Agenda

Background

From Spectral Density to the Periodogram

Spectral Estimation

Time vs. Frequency: Two Views of the Same Process

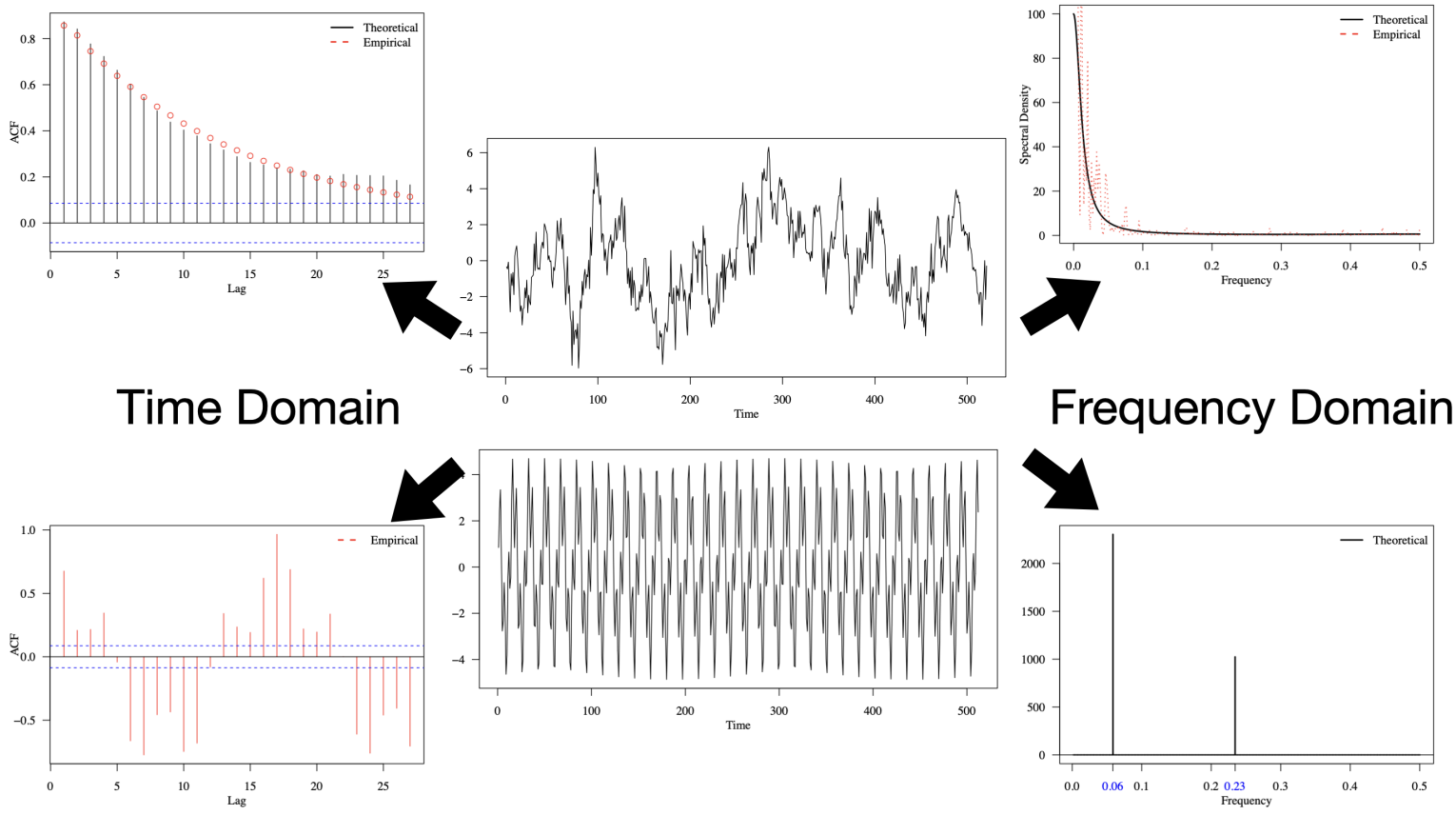
Time domain (Box & Jenkins, 1970)

- **Example (AR(1)):**
  $Y_t = \phi\, Y_{t-1} + Z_t, \quad |\phi| < 1,$
  $Z_t \sim \mathrm{WN}(0, \sigma^2)$
- **Idea:** Predict the present from the past
- **Physical view:** Dynamics in *state variables*: position $\rightarrow$ velocity $\rightarrow$ acceleration; cause $\rightarrow$ effect in time
- **Good for:** Shocks, transients, local dependencies
- **Asks:** "How does the system evolve through time?"

Frequency domain (Priestley, 1981)

- **Example (har-regression):**
  $Y_t = \alpha_0 +$
  $\sum_{j=1}^{p} \left[\alpha_{1j} \cos(\omega_j^* t) + \alpha_{2j} \sin(\omega_j^* t)\right]$
- **Idea:** Decompose into sines/cosines (*frequency*)
- **Physical view:** *Energy at resonant frequencies* (oscillators ringing at certain rates)
- **Good for:** Cycles, seasonality, filtering
- **Asks:** "Which time scales dominate the variability?"

From Autocorrelation to Spectrum: Dual Representations of Time Series



The time and frequency domains describe the same underlying process, but reveal different aspects of its structure

Waves as Building Blocks of Cyclic Time Series

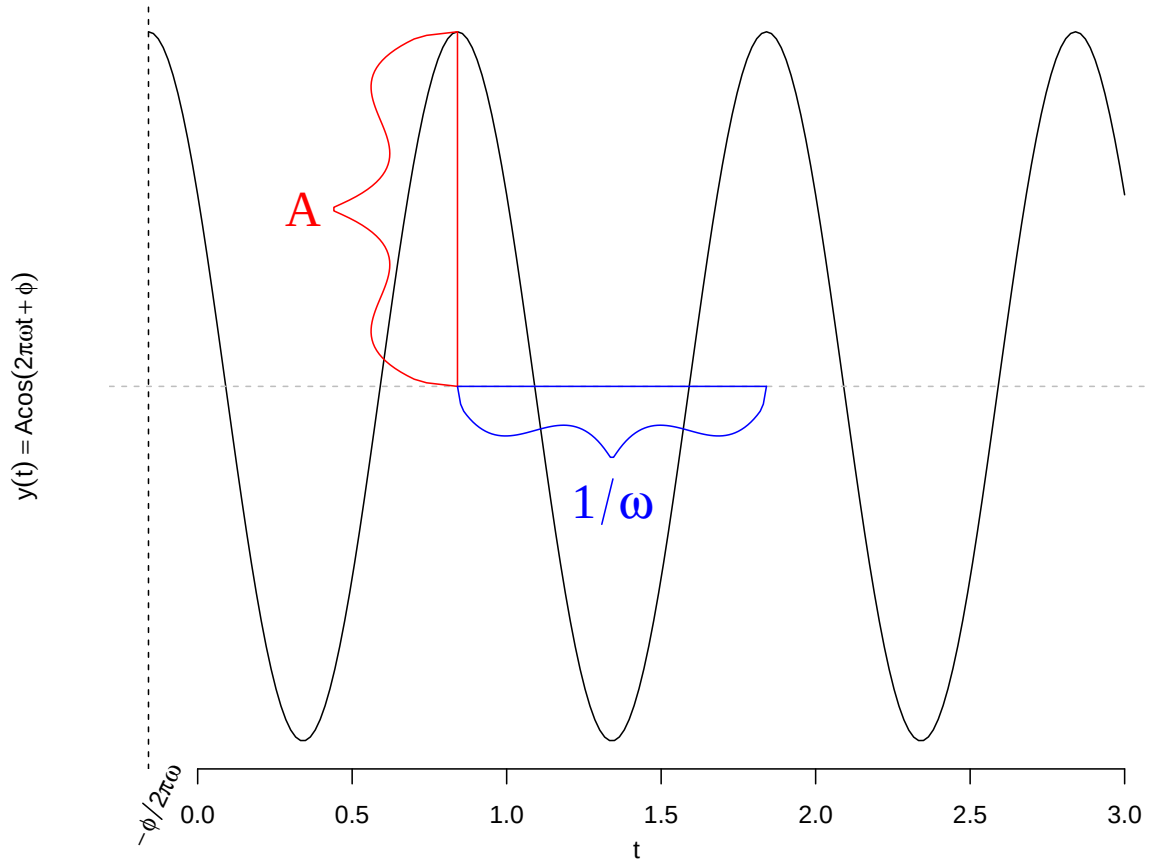
The simplest case is the cosine wave

$$\begin{aligned} Y_t &= A\cos(2\pi\omega t + \phi) \\ &= \alpha_1 \cos(2\pi\omega t) + \alpha_2 \sin(2\pi\omega t), \end{aligned}$$

$\alpha_1 = A\cos(\phi)$, $\alpha_2 = -A\sin(\phi)$, $A = \sqrt{\alpha_1^2 + \alpha_2^2}$, $\phi = \tan^{-1}\frac{-\alpha_2}{\alpha_1}$
where

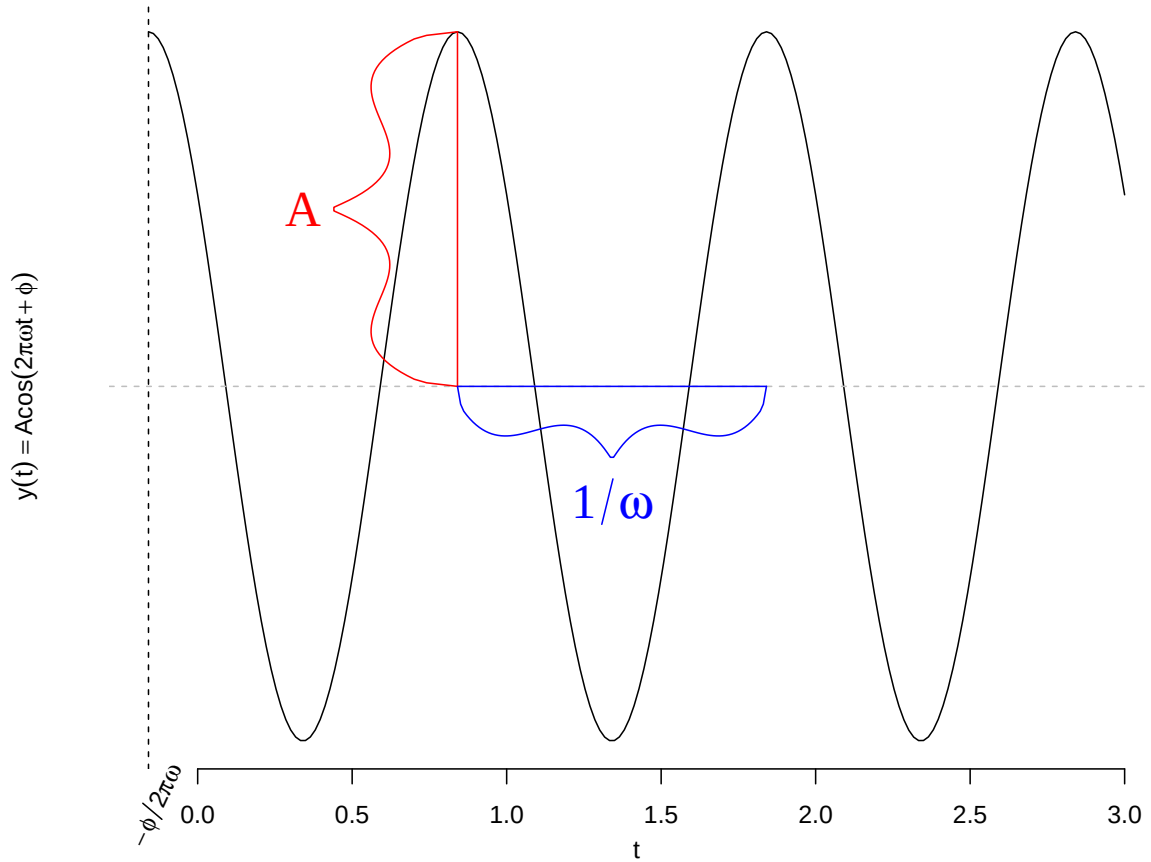
- A is amplitude: the height of the wave
- ω is frequency: how many cycles per unit time
- ϕ is phase: the start point of the cosine function

Graphical Illustration of the Cosine Wave



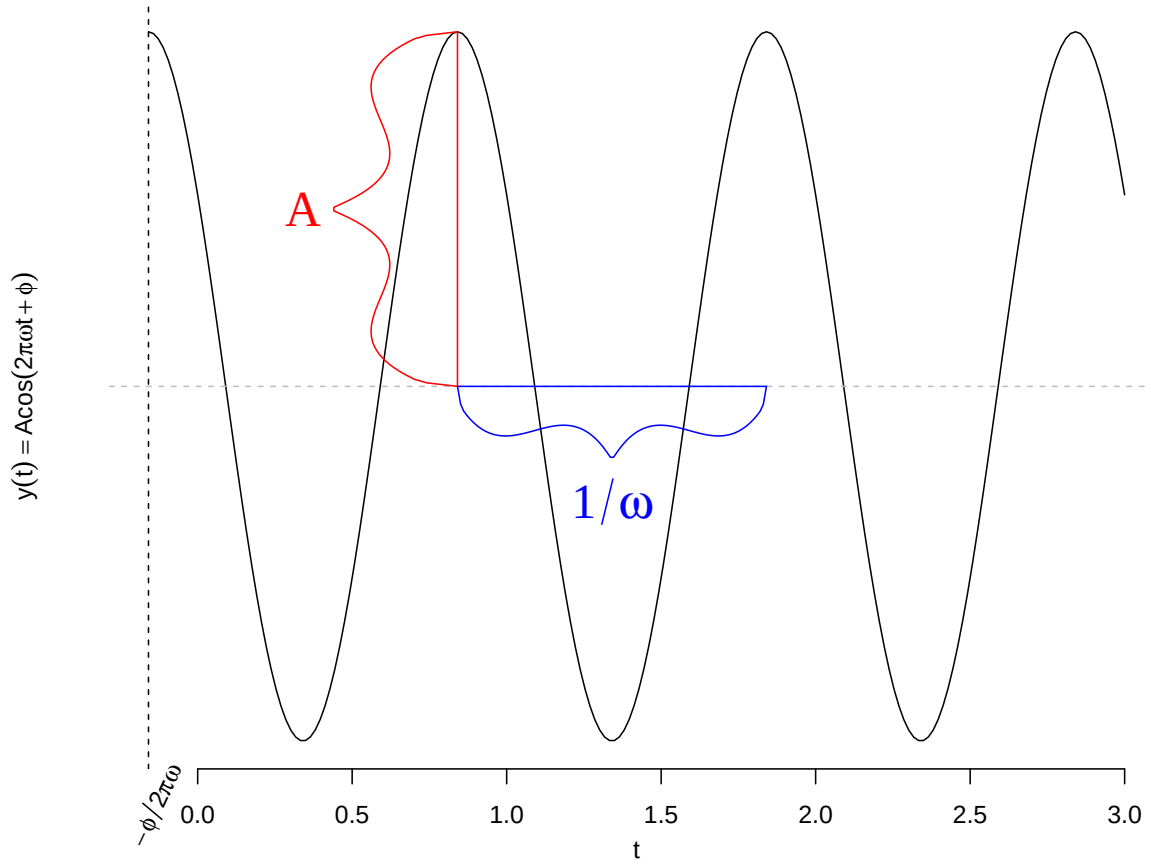
- Changing amplitude stretches vertically

Graphical Illustration of the Cosine Wave



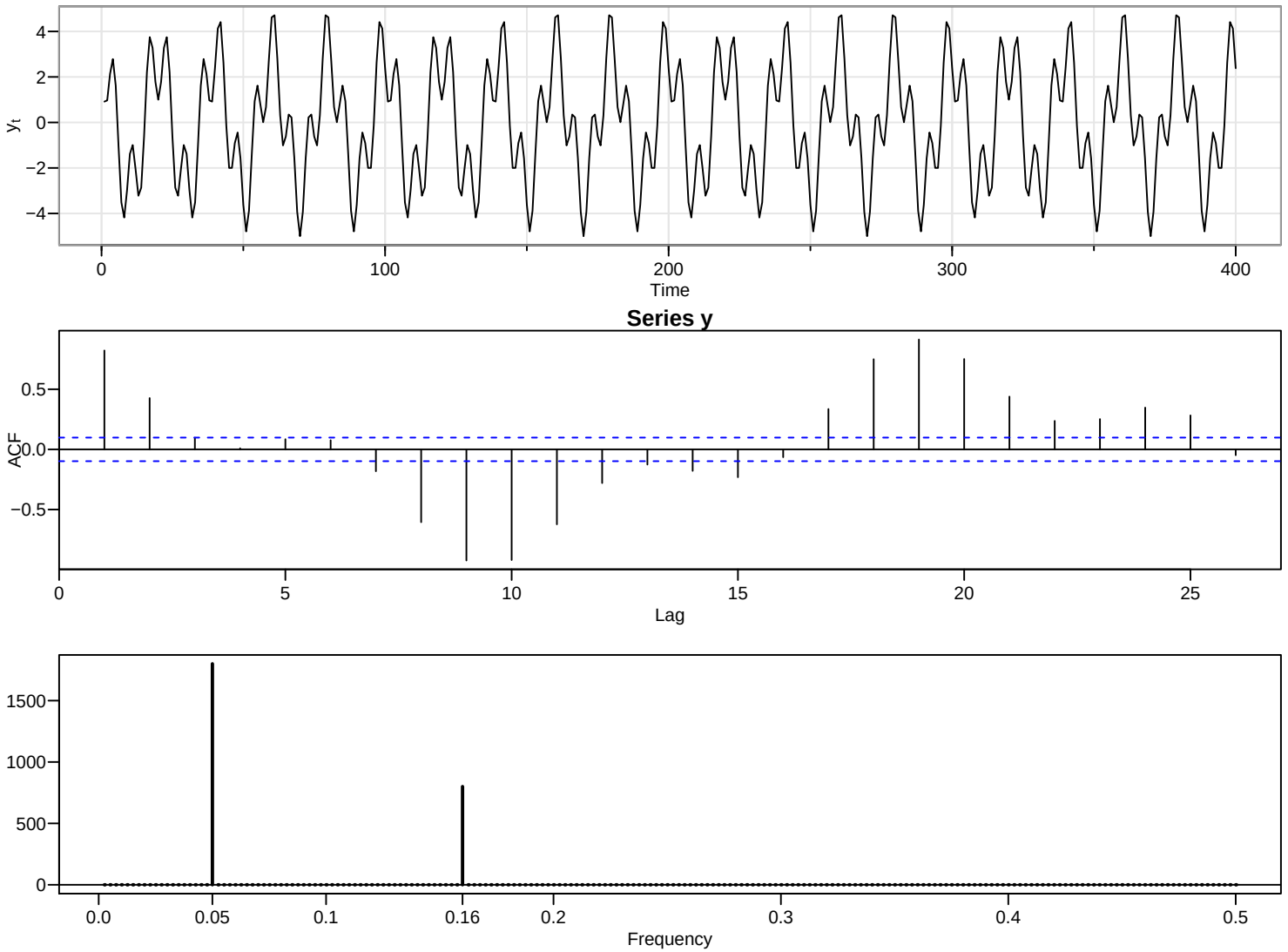
- Changing amplitude stretches vertically
- Changing frequency compresses horizontally

Graphical Illustration of the Cosine Wave

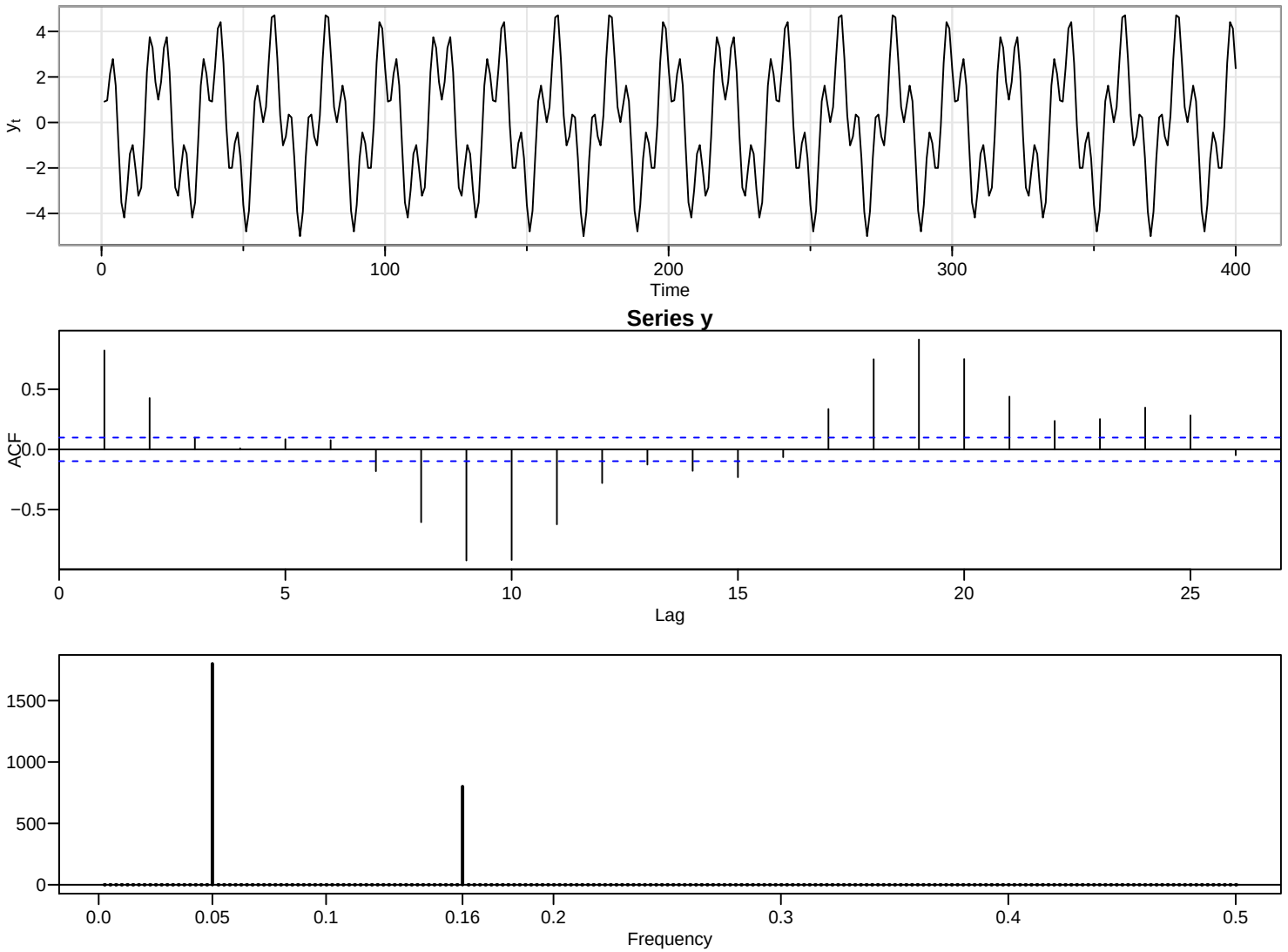


- Changing amplitude stretches vertically
- Changing frequency compresses horizontally
- Changing phase shifts horizontally

Searching Hidden Periodicities

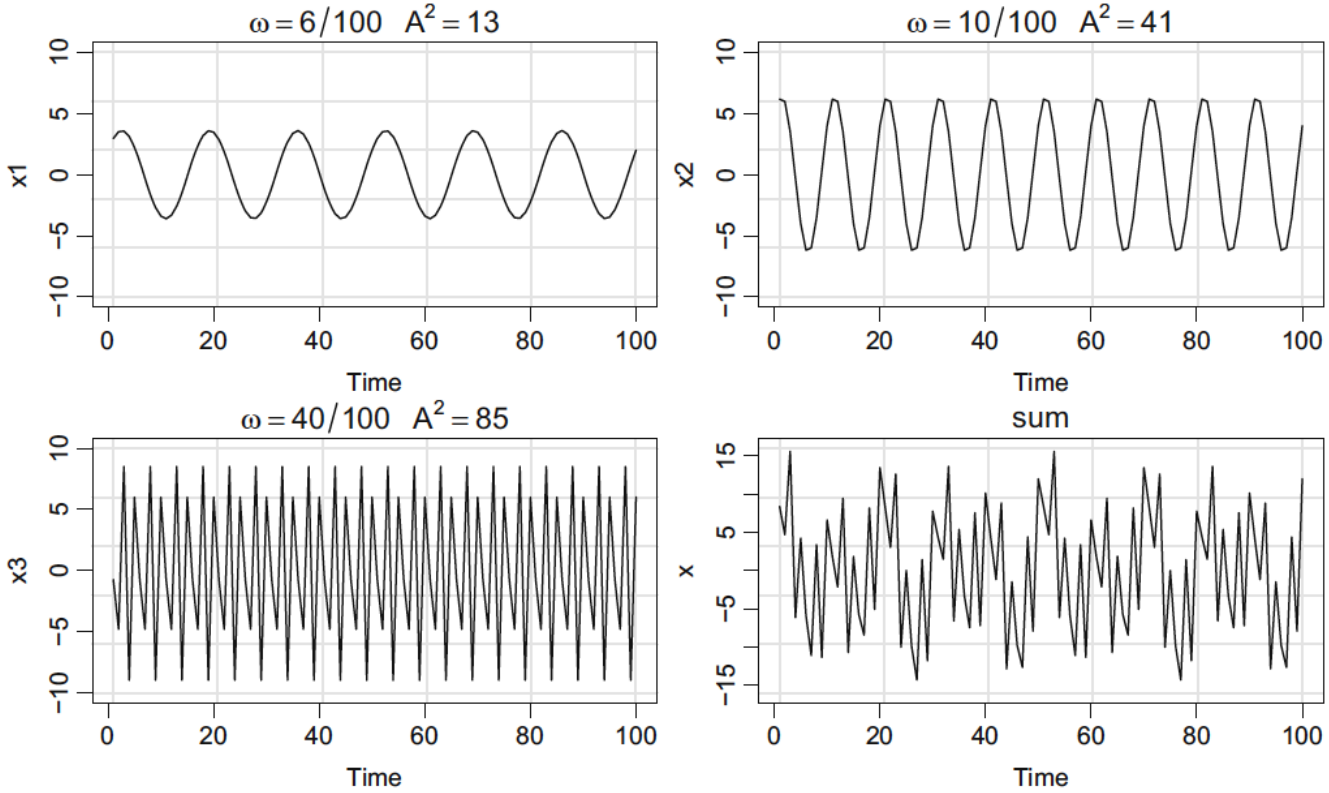


Searching Hidden Periodicities



$$y_t = 3\cos\left(2\pi(\frac{10}{200})t\right) + 2\cos\left(2\pi(\frac{32}{200}t + 0.3)\right)$$

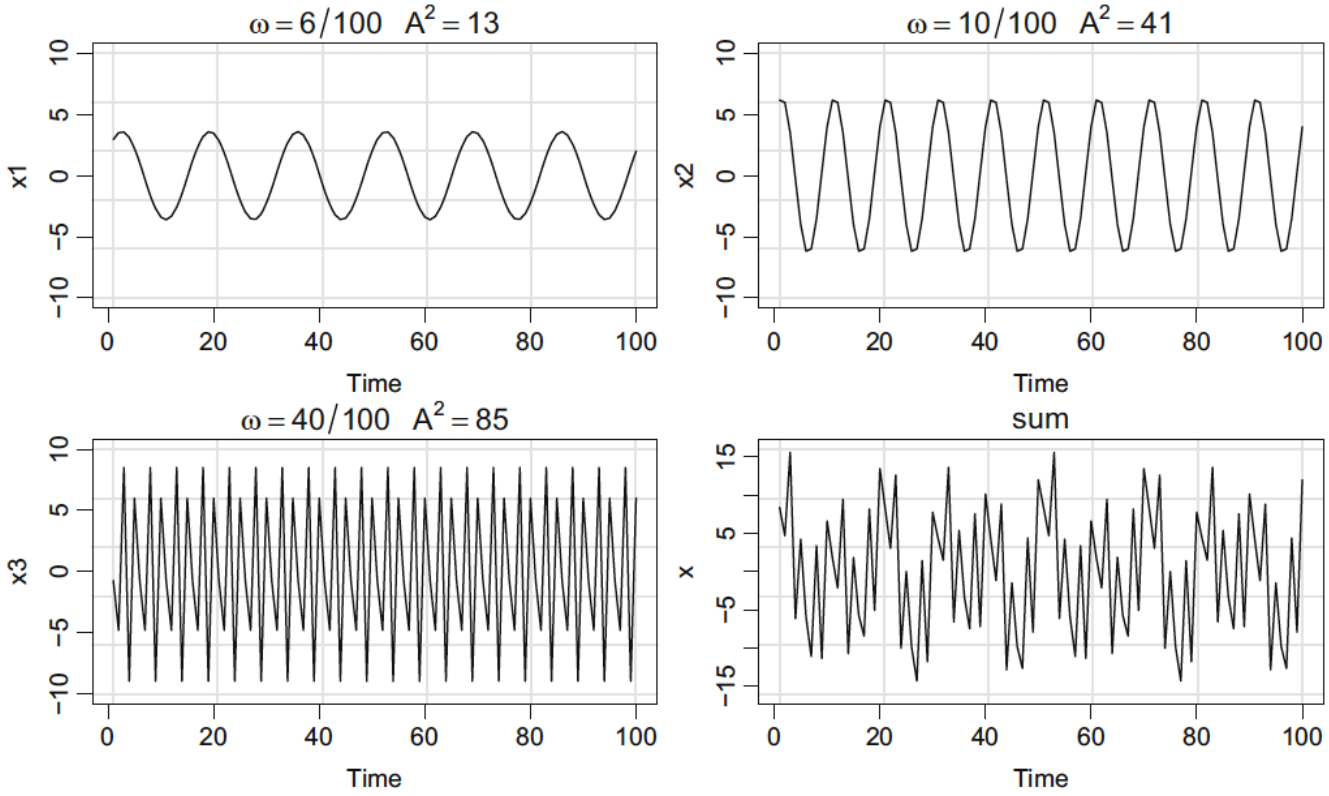
Examples of Periodic Time Series



Source: Fig. 4.1. of Shumway and Stoffer, 2017

- Combining sine and cosine waves with different frequencies and amplitudes can produce complex periodic patterns

Examples of Periodic Time Series



Source: Fig. 4.1. of Shumway and Stoffer, 2017

- Combining sine and cosine waves with different frequencies and amplitudes can produce complex periodic patterns
- **Question**: When observing a time series at discrete time points, can we distinguish all possible frequencies?

Folding Frequency and Aliasing

Let's consider $Y_{1,t} = \cos(2\pi(0.2)t)$ and $Y_{2,t} = \cos(2\pi(1.2)t)$

- At $t = 1$, $Y_{1,t} = \cos(0.4\pi t)$,
  $Y_{2,t} = \cos(2.4\pi t) = \cos(2\pi t + 0.4\pi t) = \cos(0.4\pi t) = Y_{1,t}$
- This is true for all integer values of t

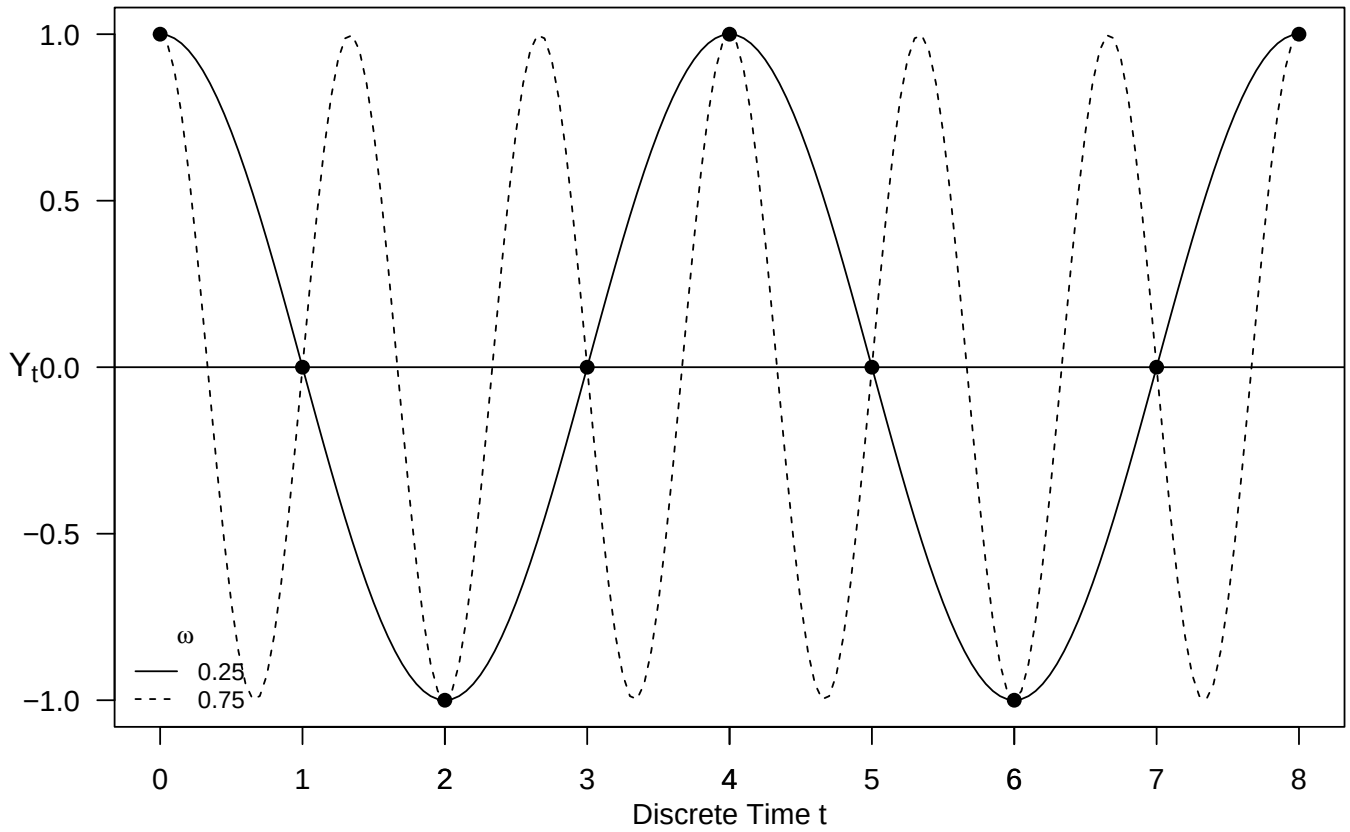
  $$\Rightarrow \omega = 1.2 \text{ is an alias of } \omega = 0.2.$$

  In general, all frequencies higher than $\omega = \frac{1}{2}$ have an alias in $0 \leq \omega \leq \frac{1}{2}$
- $\omega = \frac{1}{2}$ is the folding frequency (aka Nyquist frequency), because the shortest period that can be observed is $\frac{1}{\omega} = 2$.

Takeaway: It suffices to limit attention to $\omega \in [0, \frac{1}{2}]$

Illustration of Aliasing

$\omega = 0.25$ and $\omega = 0.75$ are aliased with one another



Takeaway: Different frequencies can produce identical observations when sampled at discrete time points. Therefore, we only need to consider frequencies between 0 and 0.5.

From Periodic Signals to Stationary Processes

Can periodic behavior coexist with stationarity? A simple way to achieve this is to introduce a random phase into a sinusoidal signal.

If

$$\begin{aligned} Y_t &= A\cos(2\pi\omega t + \phi) \\ &= \alpha_1 \cos(2\pi\omega t) + \alpha_2 \sin(2\pi\omega t), \end{aligned}$$

and ϕ is random, uniformly distributed on $[-\pi, \pi)$, then:

$$\begin{aligned} \mathbb{E}(Y_t) &= 0 \\ \mathbb{E}(Y_{t+h}Y_t) &= \frac{1}{2}A^2\cos(2\pi\omega h) \end{aligned}$$

$\Rightarrow Y_t$ is weakly stationary

A Cosine Wave With Random Phase (Continued)

Also

$$\mathbb{E}(\alpha_1) = \mathbb{E}(\alpha_2) = 0,$$
$$\mathbb{E}(\alpha_1^2) = \mathbb{E}(\alpha_2^2) = \frac{1}{2}A^2,$$
$$\text{and } \mathbb{E}(\alpha_1\alpha_2) = 0.$$

Alternatively, if the α's have these properties, then Y_t is stationary with the same mean and autocovariances:

$$\mathbb{E}(Y_t) = 0,$$
$$\mathbb{E}(Y_{t+h}Y_t) = \frac{1}{2}A^2\cos(2\pi\omega h).$$

Takeaway: Random sinusoidal coefficients provide another way to construct a weakly stationary process with oscillating autocovariance.

Representing a Periodic Process as Multiple Sines/Cosines

More generally, if $Y_t = \sum_{k=1}^{K} \left[\alpha_{k,1} \cos(2\pi\omega_k t) + \alpha_{k,2} \sin(2\pi\omega_k t)\right]$, where:

- The α's are uncorrelated with zero mean;
- $\mathrm{Var}(\alpha_{k,1}) = \mathrm{Var}(\alpha_{k,2}) = \sigma_k^2$;

then Y_t is stationary with zero mean and autocovariances

$$\gamma(h) = \sum_{k=1}^{K} \sigma_k^2 \cos(2\pi\omega_k h)$$

$$\Rightarrow \gamma(0) = \mathrm{Var}(Y_t) = \sum_{k=1}^{K} \sigma_k^2$$

For finite K, the spectrum consists of discrete frequencies rather than a continuous spectral density

From Autocovariance to Spectrum

Every weakly stationary time series Y_t with $\gamma(h)$ has a way to describe how its variance is distributed over frequencies

$$\gamma(h) = \int_{-\frac{1}{2}}^{\frac{1}{2}} e^{2\pi i\omega h} dF(\omega) = 2\int_{0}^{\frac{1}{2}} \cos(2\pi\omega h)\, dF(\omega),$$

where $F(\omega)$ is analogous to a CDF, describing the accumulated variance up to frequency ω. If $F(\omega)$ is *absolutely continuous*, it admits a spectral density function $f(\omega) = F'(\omega)$, and

$$\gamma(h) = \int_{-\frac{1}{2}}^{\frac{1}{2}} e^{2\pi i\omega h} f(\omega)\, d\omega = 2\int_{0}^{\frac{1}{2}} \cos(2\pi\omega h) f(\omega)\, d\omega$$

The autocovariance and the spectral distribution function contain the same information

From Autocovariance to Spectrum (Cont'd)

Under the absolute-summability condition $\sum_{h=-\infty}^{\infty} |\gamma(h)| < \infty$ $f(\omega)$ exists and can be written as the (uniformly convergent) Fourier series

$$f(\omega) = \sum_{h=-\infty}^{\infty} \gamma(h)\, e^{-2\pi i \omega h} = \gamma(0) + 2\sum_{h=1}^{\infty} \gamma(h)\, \cos(2\pi\omega h), \quad \omega \in \mathbb{R}.$$

Properties of the spectral density:

- $f(\omega) \geq 0$ for all ω (nonnegativity)
- $f(-\omega) = f(\omega)$ (evenness)
- $f(\omega + 1) = f(\omega)$ (1-periodicity in frequency)
- $\int_{-1/2}^{1/2} f(\omega)\, d\omega = \gamma(0) < \infty$ (total variance)

Discrete Spectral Lines vs. Continuous Spectra

Random-phase harmonic

$$Y_t = A\cos(2\pi\omega_0 t + \Phi), \Phi \sim U(-\pi, \pi).$$

$$\gamma(h) = \tfrac{A^2}{2}\cos(2\pi\omega_0 h).$$

Its spectral distribution has point masses $A^2/4$ at $\pm\omega_0$ (for $0 < \omega_0 < 1/2$).

Stationary ARMA process

$$f(\omega) = \sigma_Z^2 \frac{|\theta(e^{-2\pi i\omega})|^2}{|\phi(e^{-2\pi i\omega})|^2}.$$

Its variance is spread continuously across frequencies.

The spectral *distribution* always exists for a weakly stationary process; a density need not exist.

Example: White Noise

For white noise $\{Z_t\}$, we have seen that $\gamma(0) = \sigma_Z^2$ and $\gamma(h) = 0$ for $h \neq 0$. Thus,

$$\begin{aligned} f(\omega) &= \sum_{h=-\infty}^{\infty} \gamma(h) e^{-2\pi i \omega h} \\ &= \gamma(0) = \sigma_Z^2 \end{aligned}$$

That is, the spectral density is constant across all frequencies: each frequency contributes equally to the variance.

This is the origin of the name *white noise*: it is like white light, which is a uniform mixture of all frequencies in the visible spectrum

AR(1): Connecting Dependence and Frequency

$$Y_t = \phi Y_{t-1} + Z_t, \qquad f(\omega) = \frac{\sigma_Z^2}{1 + \phi^2 - 2\phi\cos(2\pi\omega)}.$$

- $\phi > 0$: positive persistence concentrates spectral power near frequency 0 (slow variation).
- $\phi < 0$: alternating dependence concentrates power near the Nyquist frequency $1/2$.
- $\phi = 0$: white noise has a flat spectrum.

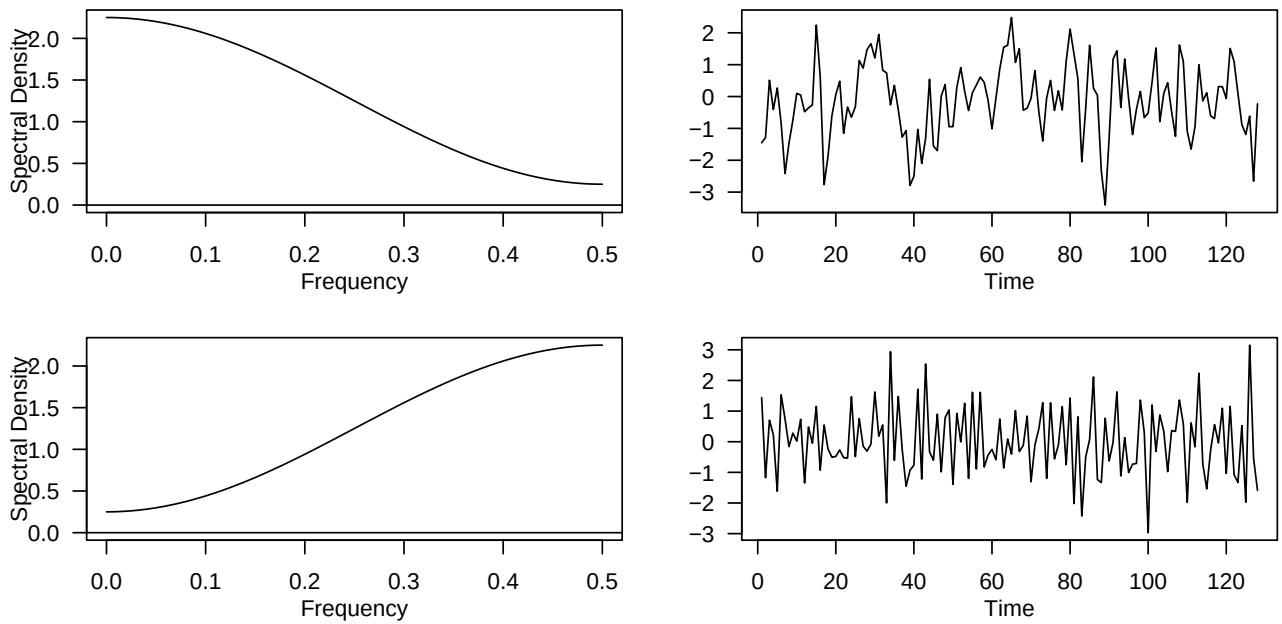
The autocovariance function and the spectral density describe the same second-order structure from different perspectives.

Examples: MA(1)

An MA(1) process $Y_t = \theta Z_{t-1} + Z_t$ is a simple filtering of white noise. Therefore, the (power) transfer function of the MA filter ($Y_t = \psi(B)Z_t$ with $\psi(z) = 1 + \theta z$) is given by:

$$\begin{aligned} |1 + \theta e^{-2\pi i\omega}|^2 &= (1 + \theta e^{-2\pi i\omega})(1 + \theta e^{2\pi i\omega}) \\ &= 1 + \theta^2 + \theta(e^{2\pi i\omega} + e^{-2\pi i\omega}) \\ &= 1 + \theta^2 + 2\theta\cos(2\pi\omega). \end{aligned}$$

Thus, we have: $f(\omega) = \left[1 + \theta^2 + 2\theta\cos(2\pi\omega)\right]\sigma_Z^2$



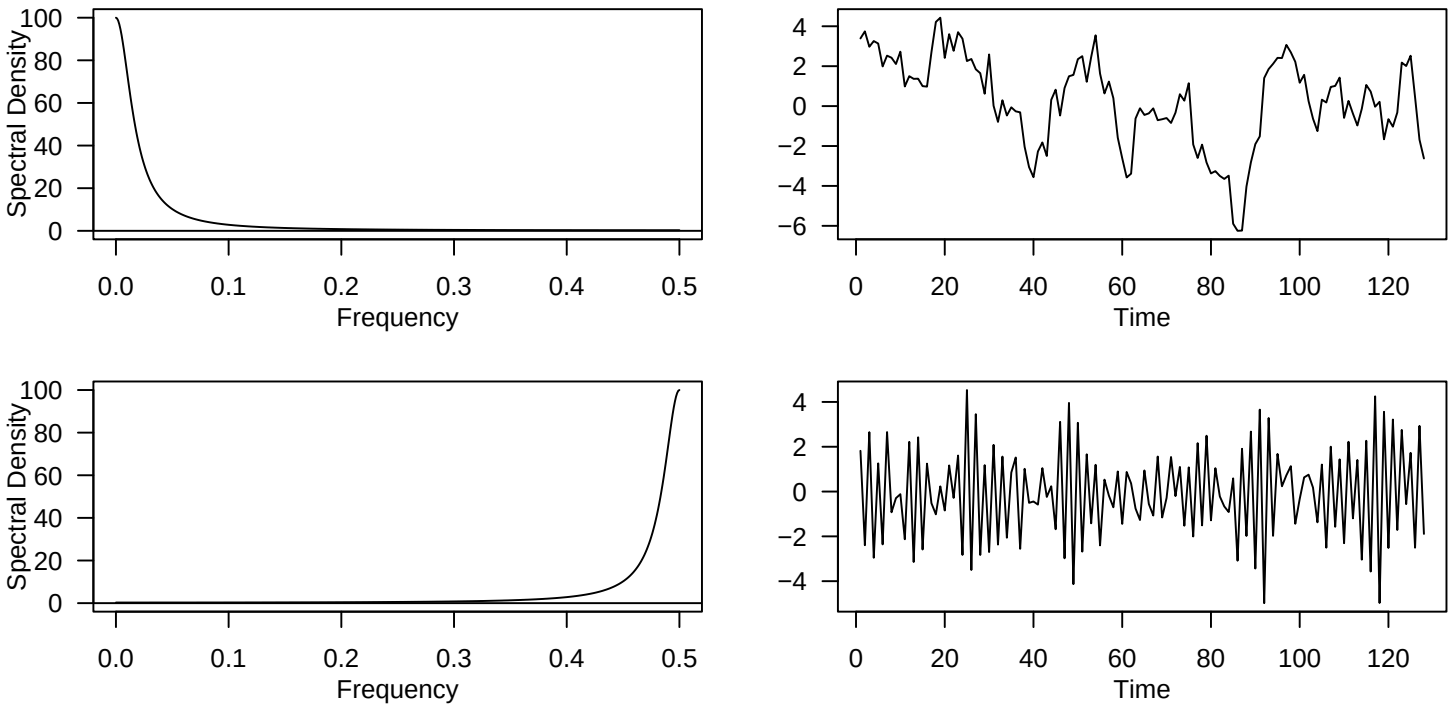
Example: AR(1)

For an AR(1) process $Y_t = \phi Y_{t-1} + Z_t$, we can write $(1 - \phi B)Y_t = Z_t \Rightarrow \psi(B) = \frac{1}{1-\phi B}$. Then we have

$$\left[1 + \phi^2 - 2\phi \cos(2\pi\omega)\right] f(\omega) = \sigma_Z^2$$

Thus,

$$f(\omega) = \frac{\sigma_Z^2}{1 + \phi^2 - 2\phi \cos(2\pi\omega)}$$



Examples: ARMA(p, q)

- For an ARMA(p, q) process:

$$\phi(B)Y_t = \theta(B)Z_t,$$

where $\phi(B)$ and $\theta(B)$ are the AR and MA polynomials, respectively.
Using the *linear filtering* result, the spectral density is:

$$f(\omega) = \sigma_Z^2 \frac{|\theta(e^{-2\pi i\omega})|^2}{|\phi(e^{-2\pi i\omega})|^2}.$$

- **Interpretation:** The spectral density shows how the variance of Y_t is distributed across frequencies. The AR and MA polynomials determine this distribution:
  - $\phi(\cdot)$: controls how strongly the series smooths or amplifies low vs. high frequencies (persistence)
  - $\theta(\cdot)$: determines short-term fluctuations or fine-scale noise

From Spectral Density to the Periodogram

Any finite time series sample $y_1, y_2, \cdots, y_n$ can be expressed as

$$y_t = \alpha_0 + \sum_{j=1}^{(n-1)/2} \left[\alpha_j \cos\left(\frac{2\pi jt}{n}\right) + \beta_j \sin\left(\frac{2\pi jt}{n}\right) \right],$$

(with one extra term if n is even)

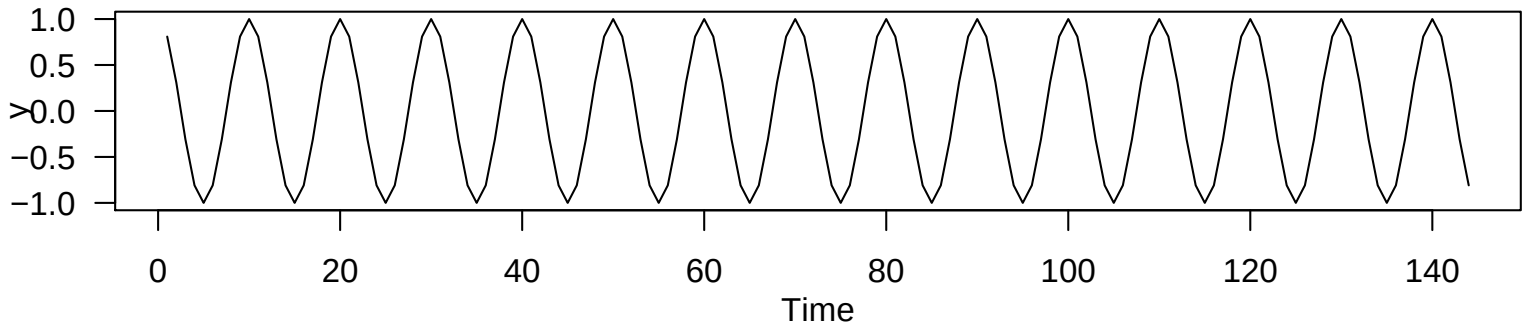
- The coefficients α_j and β_j measure how much of each frequency is present in the data
- The (scaled) periodogram

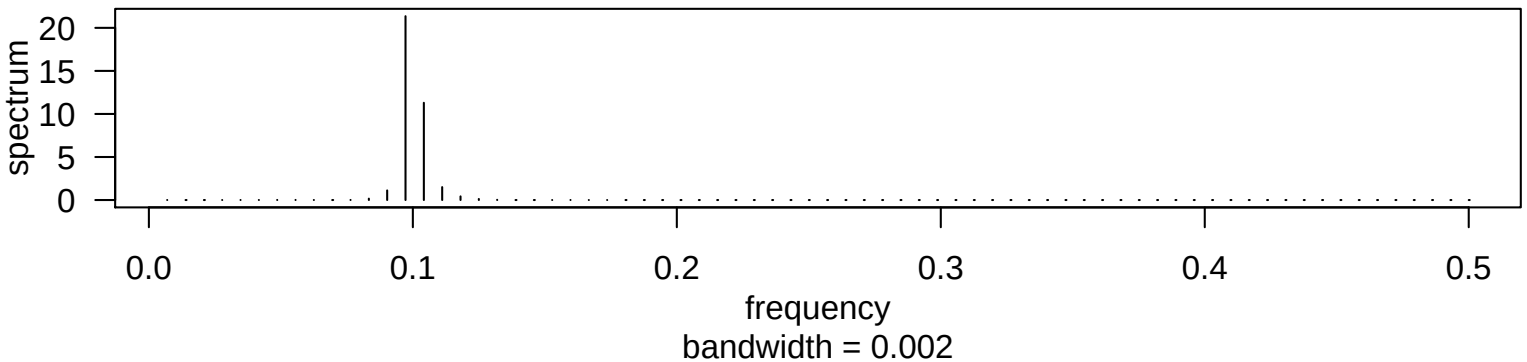
$$P(j/n) = \alpha_j^2 + \beta_j^2$$

  estimates how variance is distributed across frequencies - a sample analogue of the spectral density
- In `R`: use `spectrum()` to compute and plot $P(j/n)$

An Example: $Y_t = \cos(2\pi(0.1)t)$

```
y = cos(2 * pi * (0.1) * (1:144))
ts.plot(y)
spectrum(y, log = "no", main = "", type = "h")
```





Power around 0.1 shows the true frequency, while nearby spikes reflect spectral leakage—we'll revisit this later

Discrete Fourier Transform (DFT) and Periodogram

- The DFT transforms a time series $y_0, \ldots, y_{n-1}$ into its frequency-domain representation:

$$d(\omega_j) = \frac{1}{\sqrt{n}} \sum_{t=0}^{n-1} y_t e^{-2\pi i \omega_j t}, \quad \omega_j = \frac{j}{n},\ j = 0, \ldots, n-1.$$

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- The periodogram measures the signal's power at each frequency:

$$I(\omega_j) = |d(\omega_j)|^2.$$

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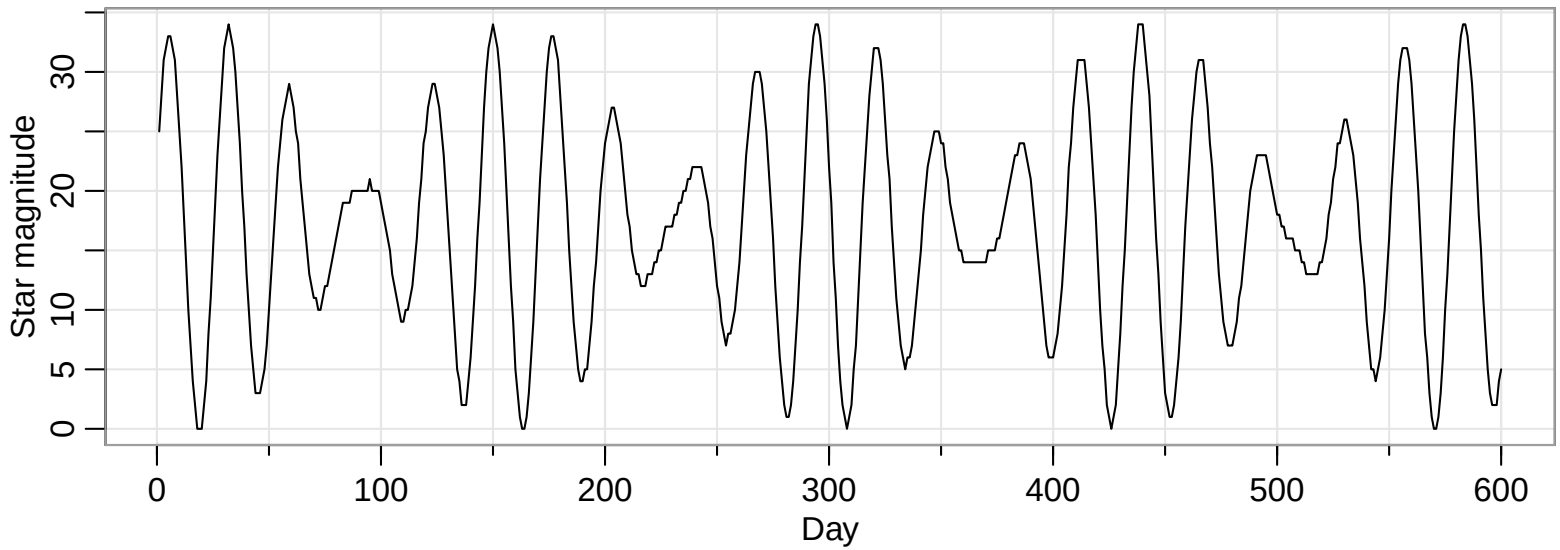
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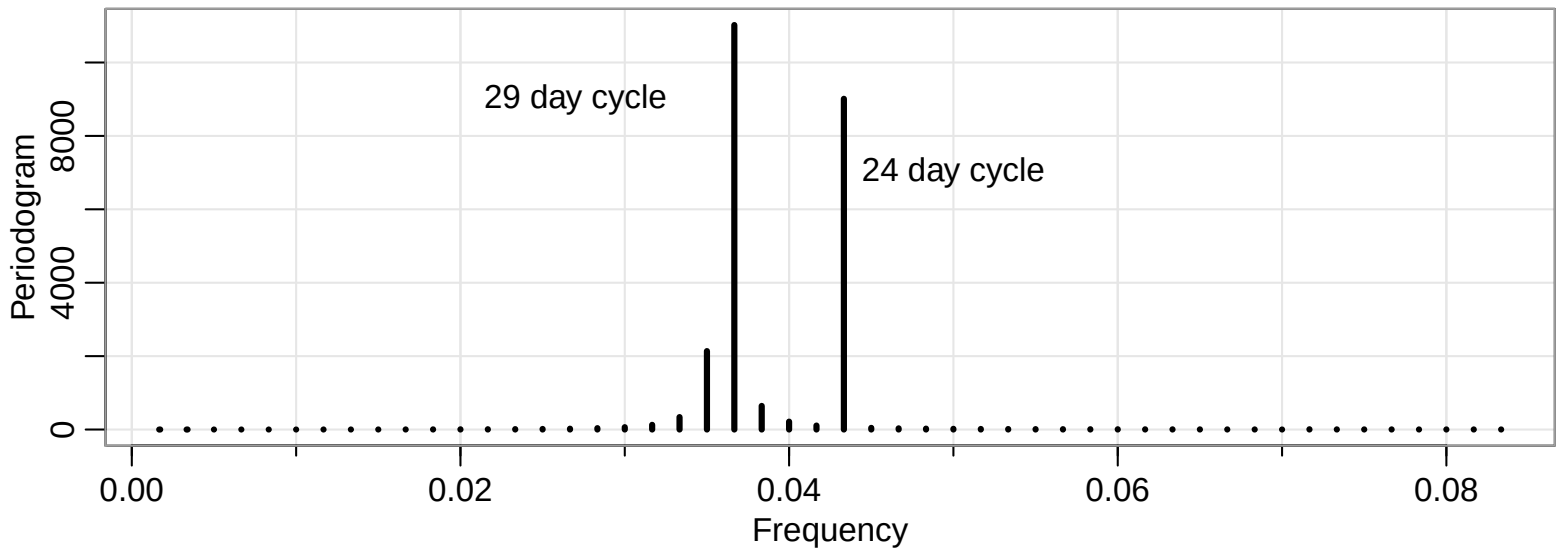
- Alternatively, in the time domain:

$$I(\omega_j) = \hat{\gamma}(0) + 2 \sum_{h=1}^{n-1} \hat{\gamma}(h) \cos(2\pi \omega_j h),$$

where $\hat{\gamma}(h)$ is the sample autocovariance at lag h

Example: Detecting Periodicities in Star Brightness [Example 4.3, SS17]





Estimating Spectral Density Using Periodogram

If n is large

$$\begin{aligned}\mathbb{E}\left[I(\omega_j)\right] &\approx \sum_{h=-(n-1)}^{n-1} \gamma(h) e^{-2\pi i \omega_j h} \\ &\approx \sum_{-\infty}^{\infty} \gamma(h) e^{-2\pi i \omega_j h} = \gamma(0) + 2\sum_{h=1}^{\infty} \gamma(h)\cos(2\pi\omega_j h) \\ &= f(\omega_j)\end{aligned}$$



- Heuristically, the spectral density is the approximate expected value of the periodogram

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- Heuristically, the spectral density is the approximate expected value of the periodogram
- Conversely, the periodogram can be used as an estimator of the spectral density
- But the periodogram values have only two degrees of freedom each, which makes it a poor estimate



DFT and Two Degrees of Freedom

For $j = 0, 1, \cdots, n-1$

$$\begin{aligned} d(\omega_j) &= n^{-\frac{1}{2}} \sum_{t=1}^{n} y_t e^{-2\pi i \omega_j t} \\ &= n^{-\frac{1}{2}} \sum_{t=1}^{n} y_t \cos(2\pi\omega_j t) - i \times n^{-\frac{1}{2}} \sum_{t=1}^{n} y_t \sin(2\pi\omega_j t) \\ &= d_{\cos}(\omega_j) - i \times d_{\sin}(\omega_j). \end{aligned}$$

- $d_{\cos}(\omega_j)$ and $d_{\sin}(\omega_j)$ are the cosine transform and sine transform, respectively, of $y_1, y_2, \cdots, y_n$
- The periodogram is $I(\omega_j) = d_{\cos}(\omega_j)^2 + d_{\sin}(\omega_j)^2$

Sampling Properties of the Periodogram

For convenience, suppose that n is odd: $n = 2m + 1$

- White noise: orthogonality properties of sines and cosines mean that
  $d_{\cos}(\omega_1), d_{\sin}(\omega_1), d_{\cos}(\omega_2), d_{\sin}(\omega_2), \cdots, d_{\cos}(\omega_m), d_{\sin}(\omega_m)$
  have zero mean, variance $\frac{\sigma_Z^2}{2}$, and are uncorrelated

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- So for Gaussian white noise

$$I(\omega_j) \sim \frac{\sigma_Z^2}{2} \times \chi_2^2$$

Sampling Distributions: General Case

General case:

$d_{\cos}(\omega_1), d_{\sin}(\omega_1), d_{\cos}(\omega_2), d_{\sin}(\omega_2), \cdots, d_{\cos}(\omega_m), d_{\sin}(\omega_m)$, have zero mean and are approximately uncorrelated, and

$$\mathrm{Var}\left[d_{\cos}(\omega_j)\right] \approx \mathrm{Var}\left[d_{\sin}(\omega_j)\right] \approx \frac{1}{2} f(\omega_j),$$

where $f(\omega_j)$ is the spectral density function

If Y_t is Gaussian,

$$\frac{I(\omega_j)}{\frac{1}{2} f(\omega_j)} = \frac{d_{\cos}(\omega_j)^2 + d_{\sin}(\omega_j)^2}{\frac{1}{2} f(\omega_j)} \approx \chi^2_2,$$

and $I(\omega_1), I(\omega_2), \cdots, I(\omega_m)$ are approximately independent

The periodogram is not a consistent estimator!

Nonparametric Spectrum Estimates

From previous slide:

$$\frac{I(\omega_j)}{\frac{1}{2}f(\omega_j)} = \frac{d_{\cos}(\omega_j)^2 + d_{\sin}(\omega_j)^2}{\frac{1}{2}f(\omega_j)} \approx \chi^2_2,$$

and $I(\omega_1), I(\omega_2), \cdots, I(\omega_m)$ are approximately independent

Problem: $I(\omega_j)$ is an approximately unbiased estimator of $f(\omega_j)$ but with too few degrees of freedom (df $= 2$) to be useful.

Specifically, $I(\omega) \dot{\sim} \frac{1}{2}f(\omega)\chi^2_2$, which implies

$$\mathbb{E}[I(\omega)] \approx f(\omega)$$

and

$$\mathrm{Var}[I(\omega)] \approx f^2(\omega)$$

Consequently, $\mathrm{Var}[I(\omega)] \overset{n\to\infty}{\neq} 0$ and thus the periodogram is not a consistent estimator of the spectral density

Why Does the Periodogram Need Smoothing?

At an interior Fourier frequency, under standard short-memory conditions,

$$I(\omega_j) \overset{\cdot}{\sim} \tfrac{1}{2} f(\omega_j) \chi^2_2.$$

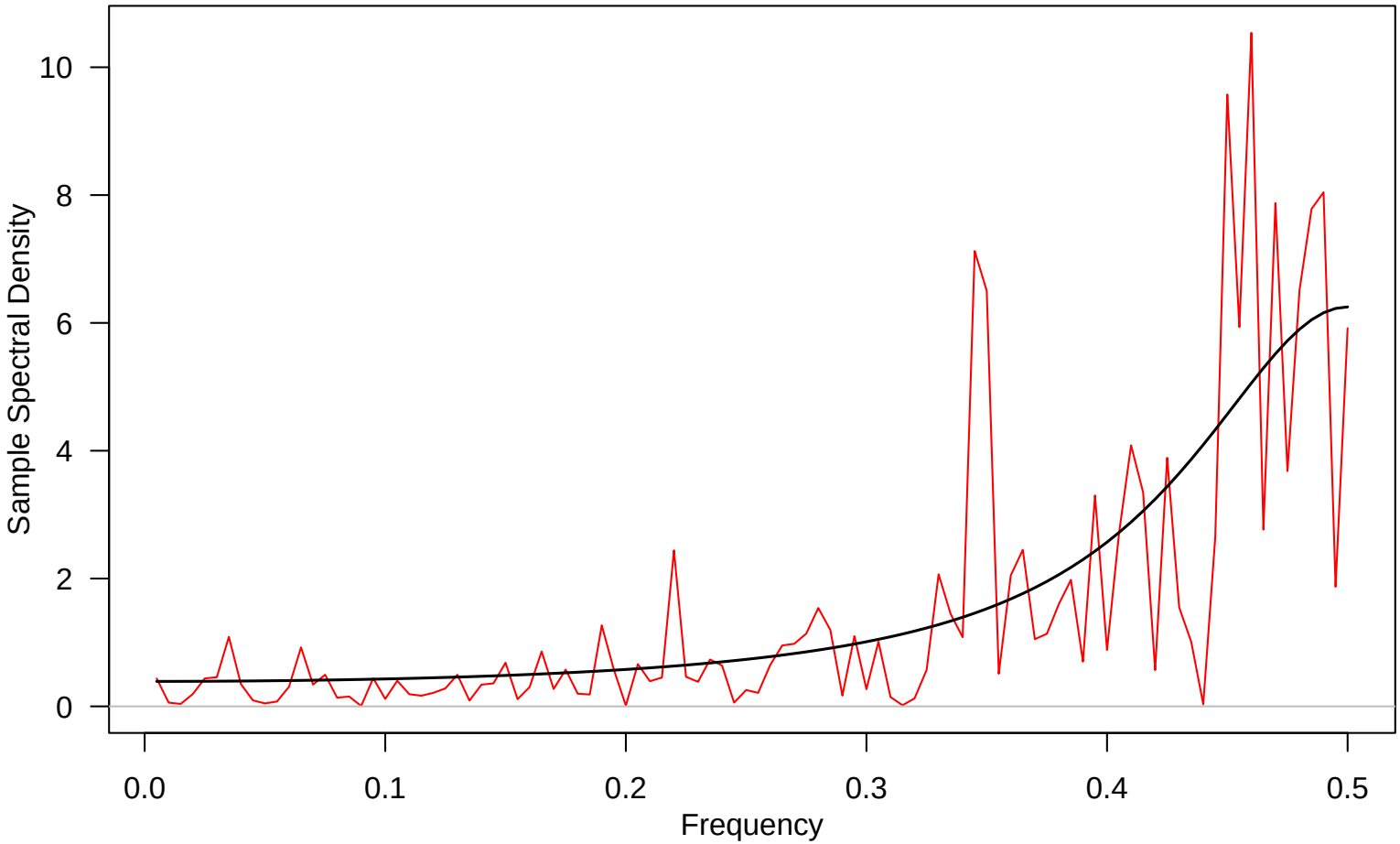
Consequently,

$$E\{I(\omega_j)\} \approx f(\omega_j), \qquad \mathrm{Var}\{I(\omega_j)\} \approx f(\omega_j)^2.$$

A longer series provides *more Fourier frequencies*, but does not eliminate the variability of a single periodogram ordinate.

Averaging neighboring ordinates reduces variance at the cost of frequency resolution.

Smoothing the Periodogram

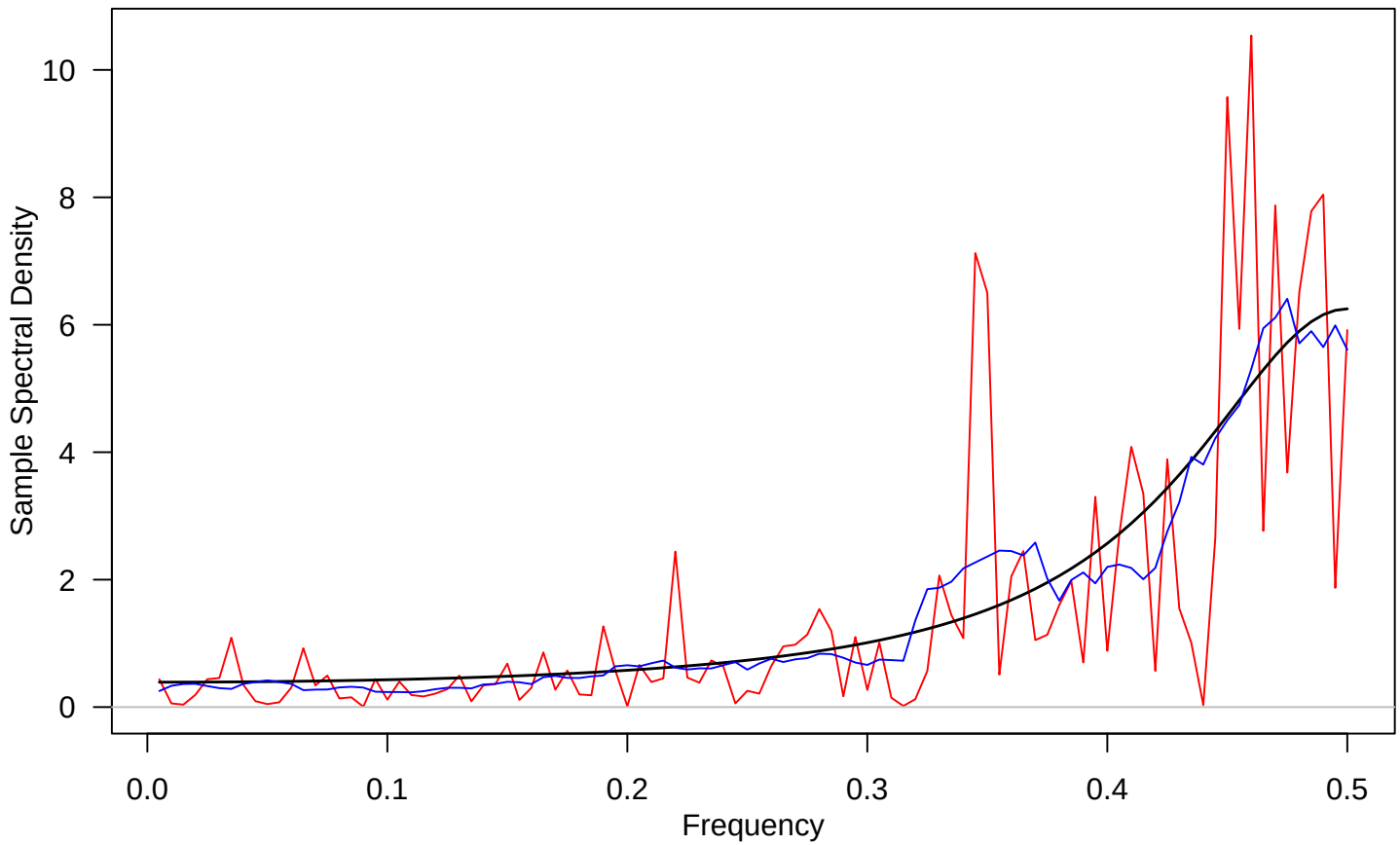


Main idea: "average" the values of the periodogram over "small" intervals of frequencies to reduce the estimation variability

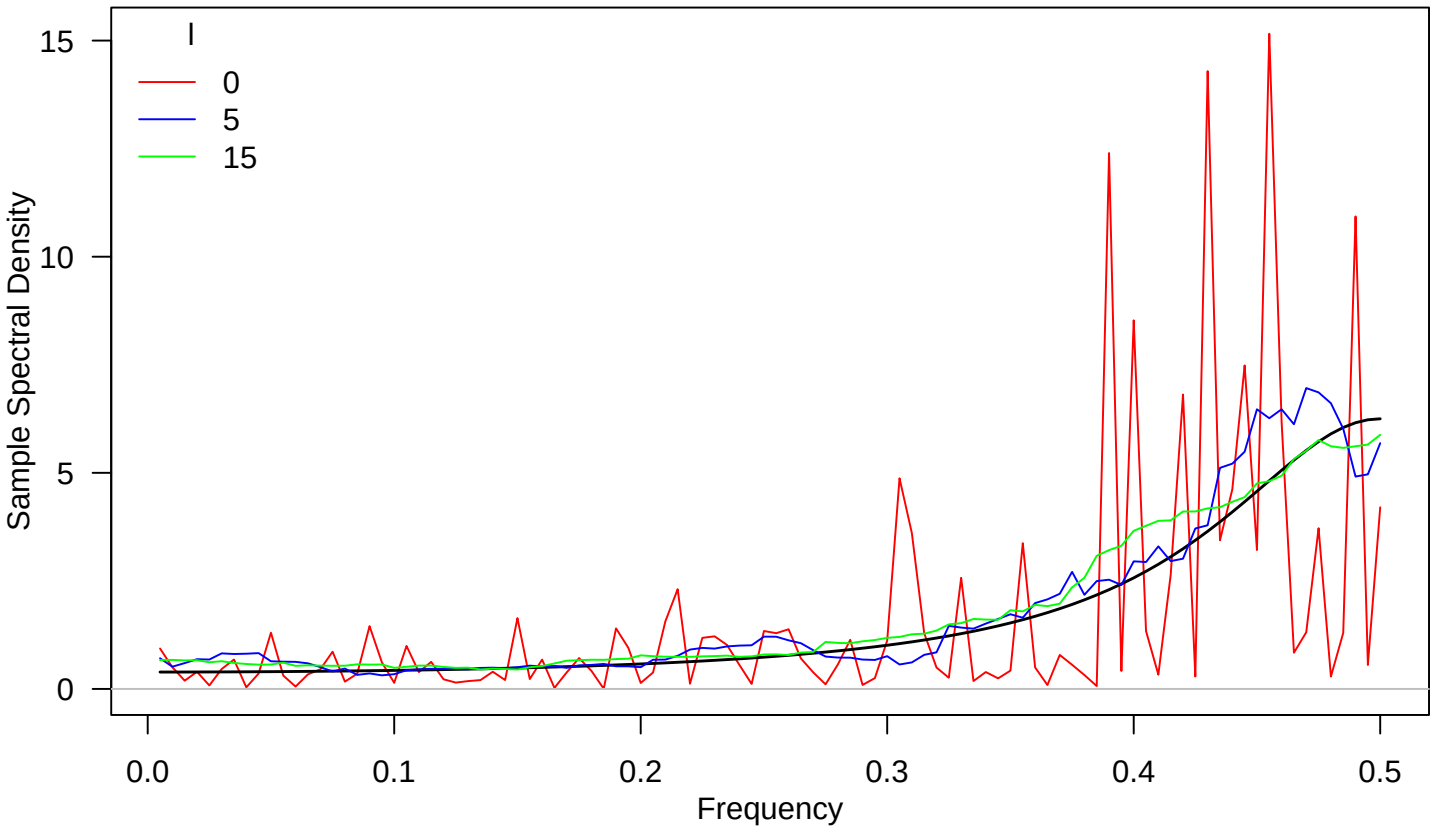
Averaged Periodogram [Daniell Spectral Window]

Use the band $[\omega_{j-l}, \omega_{j+l}]$ containing $L = 2l + 1$ Fourier frequencies:

$$\bar{f}(\omega_j) = \frac{1}{L} \sum_{k=-l}^{l} I(\omega_{j+k})$$



Tuning Parameter: l



A large l can effectively reduce the estimation variability but can also introduce bias

Choosing a Spectral Smoothing Span

- Small span: noisy estimate; narrow peaks are easier to distinguish.
- Moderate span: improved stability while retaining major peaks.
- Large span: lower variance but nearby peaks may merge.

For consistency, the number of averaged ordinates should grow while their frequency bandwidth shrinks: $l \to \infty$ and $l/n \to 0$.

Bias and Variance

Let's assume the true spectral density does not change much locally, then a Taylor expansion produces

$$\mathbb{E}[\bar{f}(\omega)] \approx \sum_{k=-l}^{l} W_l(k) f(\omega + \frac{k}{n})$$

$$\approx \sum_{k=-l}^{l} W_l(k) \left[f(\omega) + \frac{k}{n} f'(\omega) + \frac{1}{2}(\frac{k}{n})^2 f''(\omega) \right]$$

$$\approx f(\omega) + \frac{1}{n^2} \frac{f''(\omega)}{2} \sum_{k=-l}^{l} k^2 W_l(k)$$

$$\text{Bias} \approx \frac{1}{n^2} \frac{f''(\omega)}{2} \sum_{k=-l}^{l} k^2 W_l(k)$$

$$\text{Variance} \approx f^2(\omega) \sum_{k=-l}^{l} W_l^2(k)$$

Example: for Daniell rectangular spectral window, we have bias $= \frac{f''(\omega)}{2n^2} \frac{l(l+1)}{3}$ and variance $\frac{f^2(\omega)}{2l+1}$

Pointwise Confidence Intervals for $f(\omega)$

The distribution of $\frac{\nu \bar{f}(\omega)}{f(\omega)}$ can be approximated by $\chi^2_{df=\nu}$, where

$$\nu = \frac{2}{\sum_{k=-l}^{l} W_l^2(k)}$$

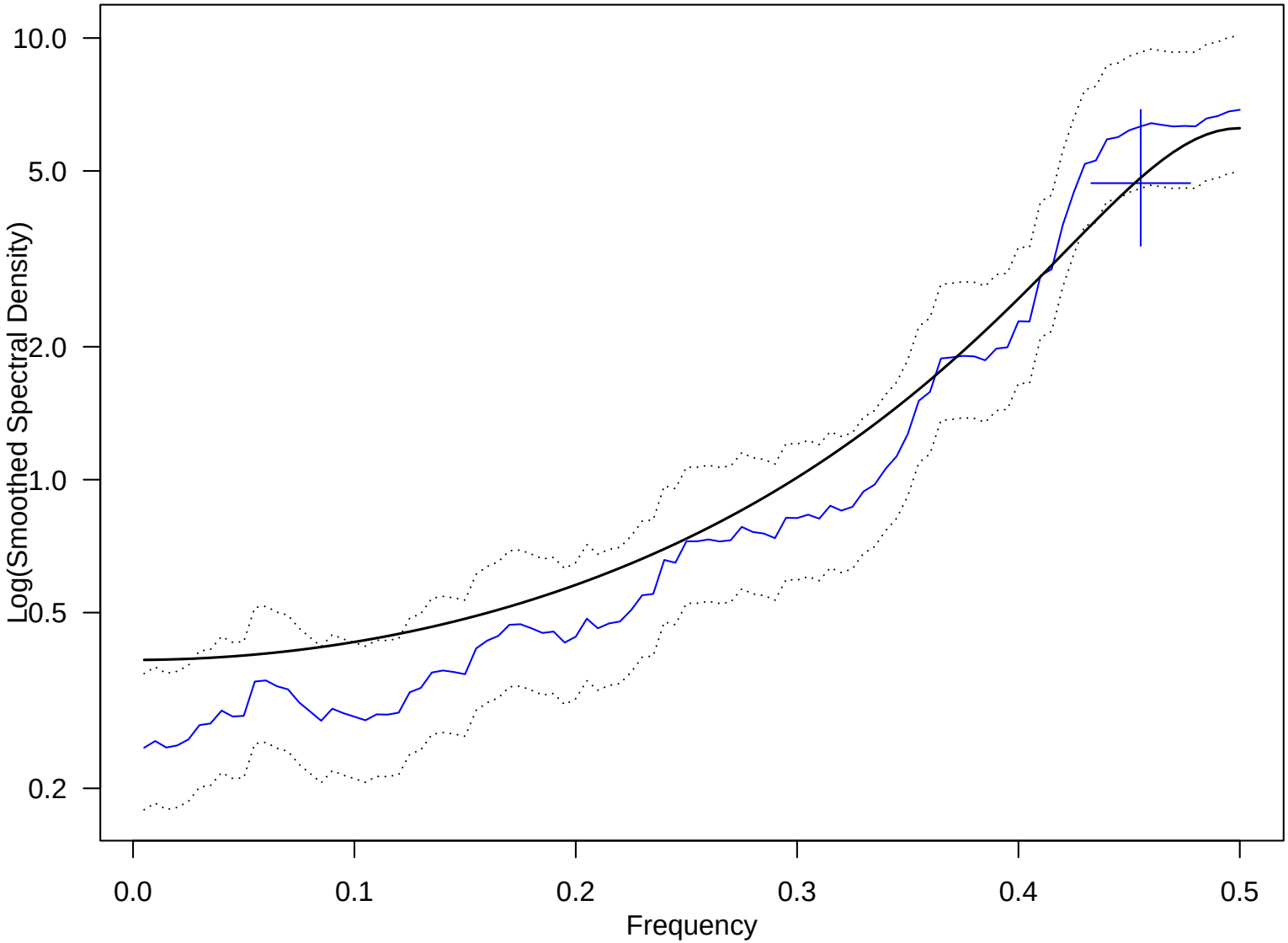
$\Rightarrow 100(1-\alpha)\%$ CI for $f(\omega)$

$$\frac{\nu \bar{f}(\omega)}{\chi^2_{df=\nu,1-\frac{\alpha}{2}}} < f(\omega) < \frac{\nu \bar{f}(\omega)}{\chi^2_{df=\nu,\frac{\alpha}{2}}}$$

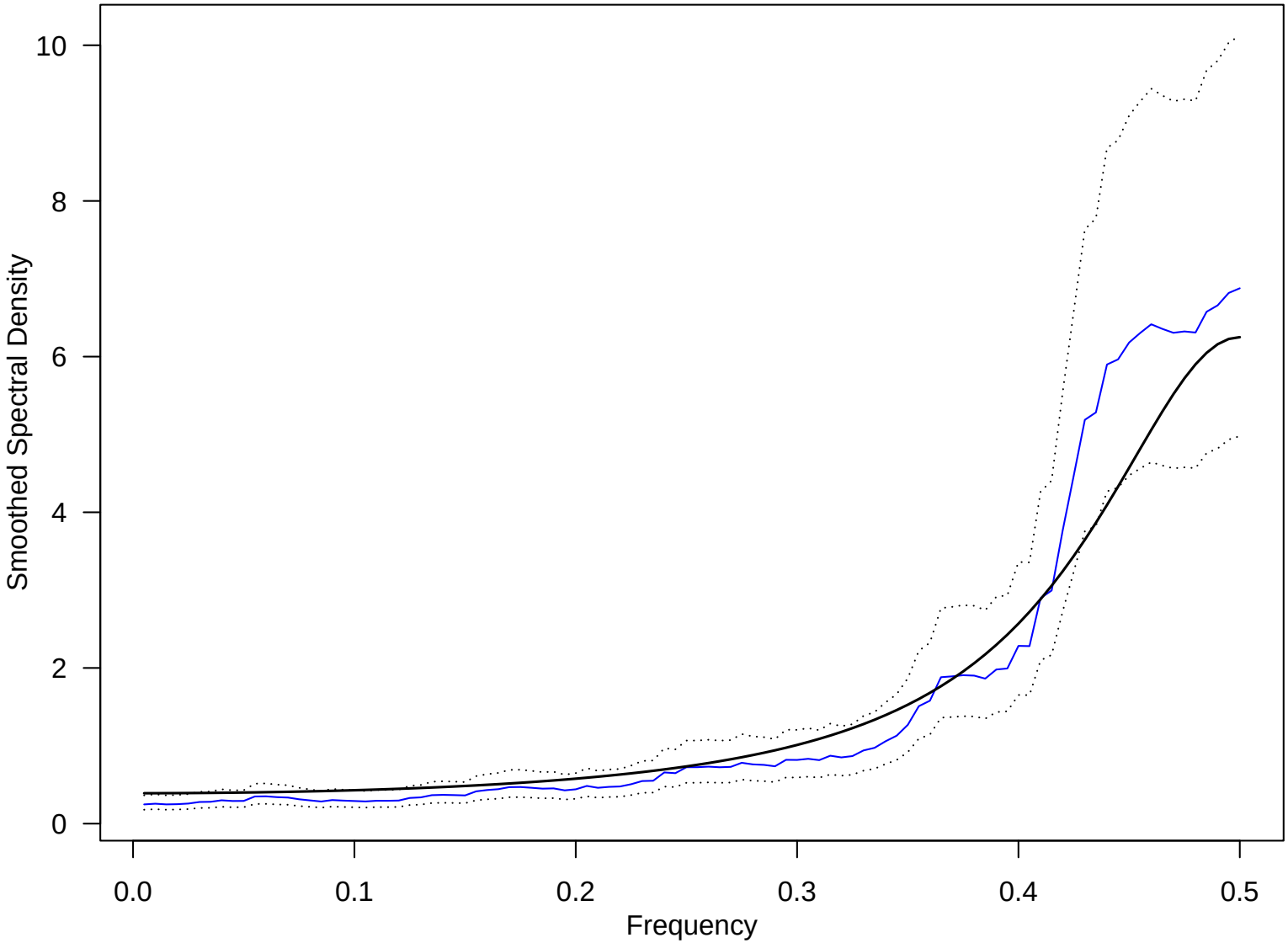
Taking logs we obtain an interval for the logged spectrum:

$$\log[\bar{f}(\omega)] + \log\left[\frac{\nu}{\chi^2_{\nu,1-\frac{\alpha}{2}}}\right] < \log[f(\omega)] < \log[\bar{f}(\omega)] + \log\left[\frac{\nu}{\chi^2_{\nu,\frac{\alpha}{2}}}\right]$$

Pointwise Confidence Intervals for $f(\omega)$: Log-Scale



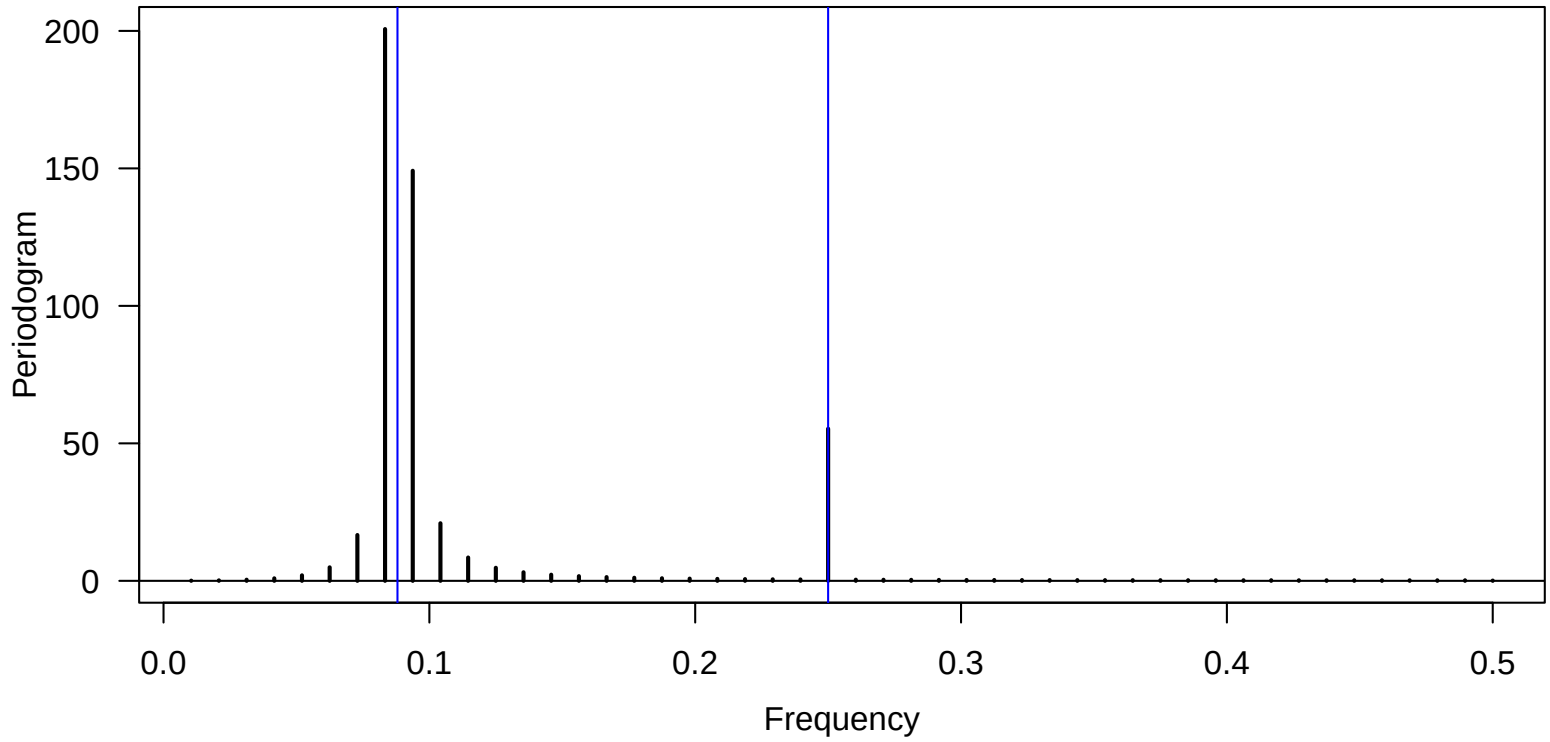
Pointwise Confidence Intervals for $f(\omega)$: Original Scale



Spectral Leakage

Much of the previous discussion has assumed that the frequencies of interest are the Fourier frequencies, i.e., $\omega_j = \frac{j}{n}$. What happens if that is not the case?

Example: $Y_t = 3\cos(2\pi(0.088)t) + \sin(2\pi(\frac{24}{96})t), \quad t = 1, \cdots, 96$

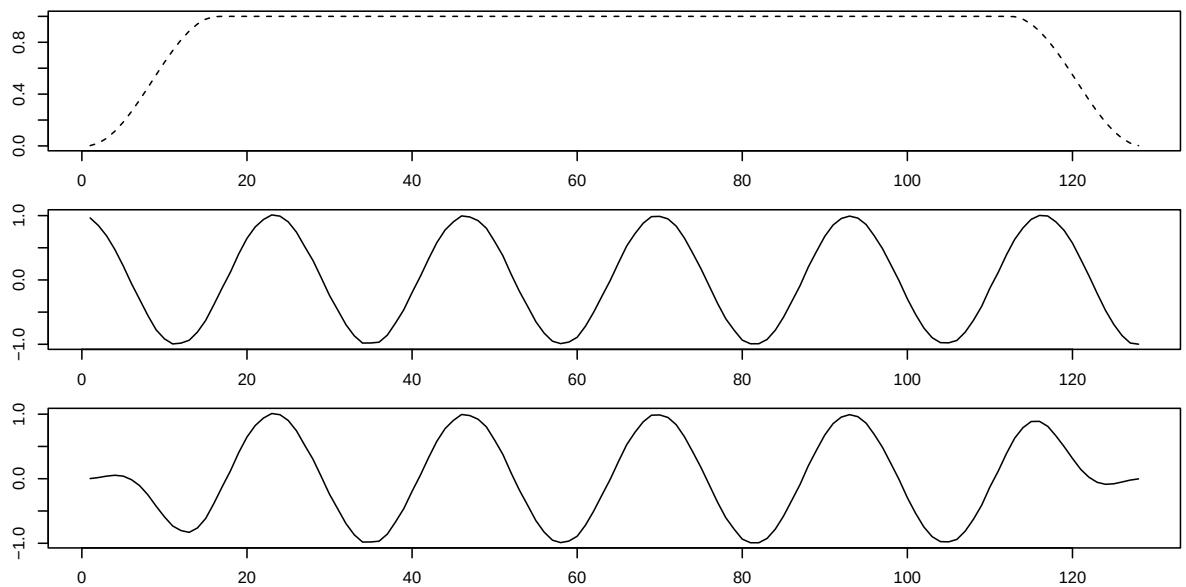


Power at non-Fourier frequencies will leak into the nearby Fourier frequencies

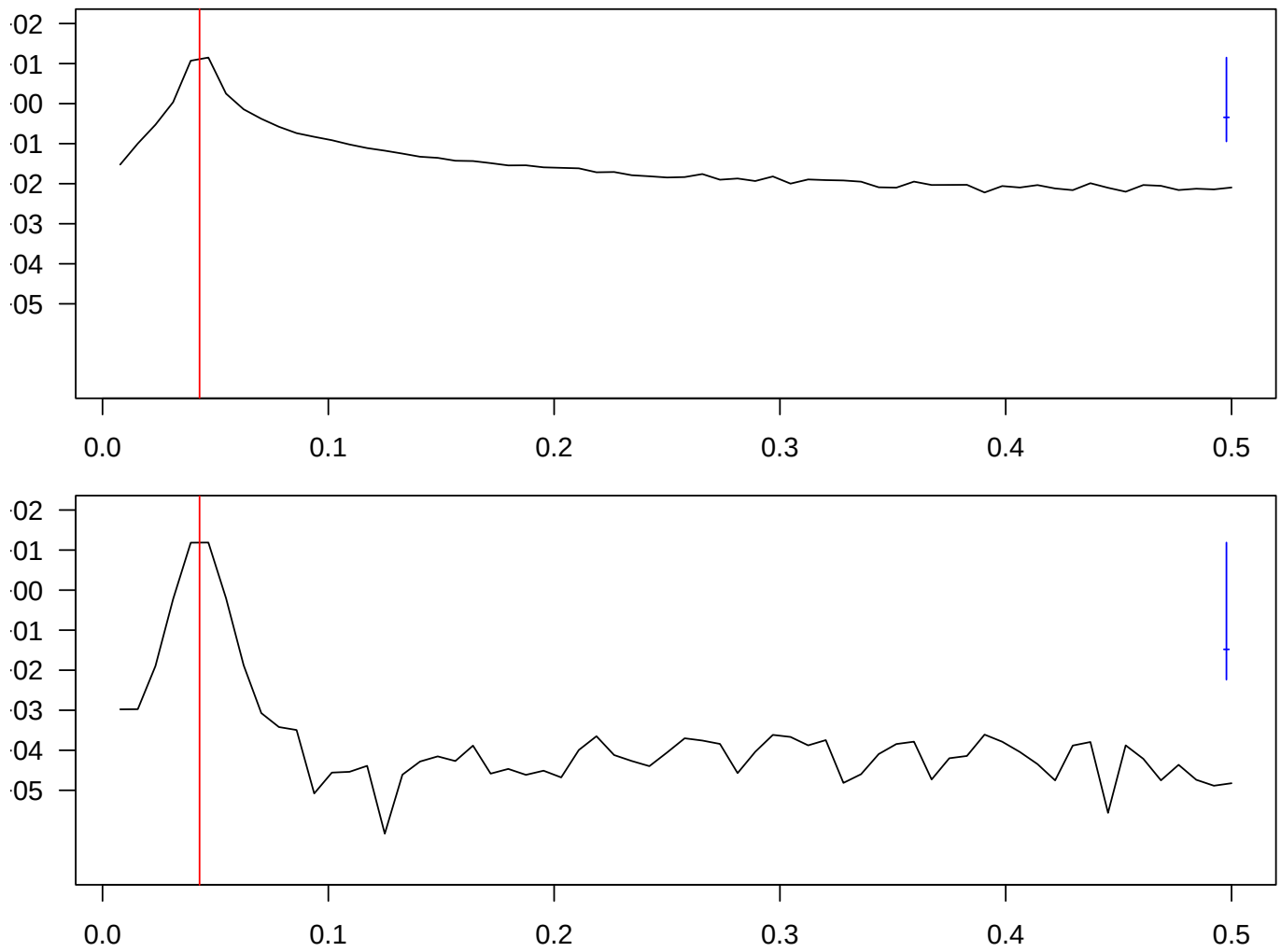
Tapering

Tapering is one method used to alleviate the issue of spectral leakage, where power at non-Fourier frequencies leaks into the nearby Fourier frequencies

Main idea: replace the original series by the tapered series, i.e., $\tilde{y}_t = h_t y_t$. Tapers h_t's generally have a shape that enhances the center of the data relative to the extremities to reduce the end effects of computing a Fourier transform on a series of finite length



Effect of Tapering



Tapering reduces leakage at the cost of frequency resolution and potentially greater uncertainty in spectral estimation

Leakage, Tapering, and Smoothing: Different Roles

- **Spectral leakage:** Finite observation windows cause spectral spreading through convolution with the spectral window, introducing bias.
- **Tapering:** Downweights observations near the boundaries to suppress sidelobes, reducing distant leakage at the cost of a wider main lobe.
- **Smoothing:** Averages neighboring periodogram ordinates to reduce estimation variance, at the cost of increased spectral bias and reduced resolution.

Key distinction: Tapering modifies the spectral window; smoothing reduces estimation variance. Both affect spectral bias and frequency resolution.